

► Simple Types

$$\alpha ::= \iota \mid o \mid \alpha_1 \rightarrow \alpha_2$$

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Individuals

Booleans (True and False)

Functions

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Possible worlds

Individuals

Booleans (True and False)

Functions

► Simple Types

$$\alpha ::= \iota \mid \mu \mid o \mid \alpha_1 \rightarrow \alpha_2$$

► HOL Syntax

$$\begin{aligned} s, t &::= C_\alpha \mid x_\alpha \\ &\mid (\lambda x_\alpha. s_\beta)_{\alpha \rightarrow \beta} \mid (s_{\alpha \rightarrow \beta} t_\alpha)_\beta \\ &\mid (\neg_{o \rightarrow o} s_o)_o \mid (s_o \vee_{o \rightarrow o \rightarrow o} t_o)_o \mid (\forall x_\alpha. t_o)_o \end{aligned}$$

Constant Symbols

Variable Symbols

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Constant Symbols

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Abstraction

Application

- ▶ Simple Types
- ▶ HOL Syntax

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Constant Symbols

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Logical Connectives

- ▶ Simple Types
- ▶ HOL Syntax

$$\alpha ::= \iota \mid \mu \mid o \mid \alpha_1 \rightarrow \alpha_2$$

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No Binder for Quantification needed!
Use constant Π together with λ !

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► HOL is (meanwhile) well understood

- Origin

[Church, J.Symb.Log., 1940]

- Henkin-Semantics

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[Andrews, J.Symb.Log., 1971, 1972]

- Extens./Intens.

[BenzmüllerBrownKohlhase, J.Symb.Log., 2004]

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► HOL with Henkin-Semantics:

semi-decidable & compact

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► HOL with Henkin-Semantics: **semi-decidable & compact**

► ATPs: **LEO-II, Satallax**, TPS, Isabelle, Nitpick, Refute

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For our experiments only LEO-II and Satallax were selected.
However, the overall approach works for all these ATPs.

[Muskens, J.Symb.Log., 2007]

► HOL with Henkin-Semantics: semi-decidable & compact

► ATPs: LEO-II, Satallax, TPS, Isabelle, Nitpick, Refute

Kripke style semantics

TERMS2HOL: $t \longrightarrow t_\mu$

$M, g, s \models P(x^1, \dots, x^n)$	iff	$\langle g(x^1), \dots, g(x^n) \rangle \in I(P, s)$
$M, g, s \models \neg \varphi$	iff	not $M, g, s \models \varphi$
$M, g, s \models \varphi \vee \psi$	iff	$M, g, s \models \varphi$ or $M, g, s \models \psi$
$M, g, s \models \Box \varphi$	iff	$M, g, u \models \varphi$ for all u with $R(s, u)$
$M, g, s \models \forall x. \varphi$	iff	$M, [d/x]g, s \models \varphi$ for all $d \in D$

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Semantic embedding:

FML2HOL: $\varphi \longrightarrow \varphi_\rho$ with type $\rho := \iota \rightarrow o$

$P(x^1, \dots, x^n)$	=	$(P_{\mu^n \rightarrow \rho} x_\mu^1 \dots x_\mu^n)$
$\neg$	=	$\lambda \varphi_\rho. \lambda w_\iota. \neg(\varphi w)$
$\vee$	=	$\lambda \varphi_\rho. \lambda \psi_\rho. \lambda w_\iota. (\varphi w) \vee (\psi w)$
$\Box$	=	$\lambda \varphi_\rho. \lambda w_\iota. \forall v_\iota. \neg(R w v) \vee (\varphi v)$
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→ Soundness & Completeness: [BenzmüllerPaulson, LogicaUniversalis, 2012]

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Completeness: [BenzmüllerPaulson, LogicaUniversalis, 2012]

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FML in HOL — Running Example —

$$(\Diamond \exists x Pfx \wedge \Box \forall y (\Diamond Py \Rightarrow Qy)) \Rightarrow \Diamond \exists z Qz$$

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$$\Box = \lambda \varphi_p. \lambda w_l. \forall v_l. \neg (Rwlv) \vee (\varphi v)$$

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...

$$\forall w (\neg \neg (\neg \neg \forall v (\neg R w v \vee \neg \neg \forall x \neg P(fx) v) \vee \neg \forall v (\neg R w v \vee \forall y (\neg \neg \forall u (\neg R v u \vee \neg P y u) \vee Q y v))) \vee \neg \forall v (\neg R w v \vee \neg \neg \forall z \neg Q z v))$$

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Axiomatization of properties of accessibility relation R

Logic K: no axioms

Logic T: (*reflexive* R) — which expands into $\forall x. R_{xx}$

Logic S4: (*reflexive* R) $\wedge$ (*symmetric* R) $\wedge$ (*transitive* R)

Logic ...

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Logic

This automates FML with **constant domain semantics** in HOL

To obtain **varying domain semantics**:

- ▶ modify quantifier: $\sqcap = \lambda q_{\mu \rightarrow \rho} \lambda w_{\iota} \forall x_{\mu} \text{ExistsInW}_{xw} \Rightarrow q_{xw}$
- ▶ add non-emptiness axiom: $\forall w_{\iota} \exists x_{\mu} \text{ExistsInW}_{xw}$
- ▶ add designation axioms for constants c : $\forall w_{\iota} \text{ExistsInW}_{cw}$
(similar for function symbols)

To obtain **varying domain semantics**:

- ▶ modify quantifier: $\sqcap = \lambda q_{\mu \rightarrow \rho} \lambda w_t \forall x_\mu \text{ExistsInW}_{xw} \Rightarrow q_{xw}$
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To obtain **cumulative domain semantics**:

- ▶ add axiom: $\forall x_\mu \forall v_t \forall w_t \text{ExistsInW}_{xv} \wedge R_{vw} \Rightarrow \text{ExistsInW}_{xw}$

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- ▶ modify quantifier: $\Box = \lambda q_{\mu \rightarrow \rho} \lambda w_t \forall x_\mu \text{ExistsInW}_{xw} \Rightarrow q_{xw}$
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The FML2THF-tool

- ▶ Syntax mapping and provision of axioms wrt.
 - ▶ selected logic: K, D, T, S4, S5, etc.
 - ▶ selected domain condition: varying, cumulative, constant

Experiments using QMLTP Library

Number of proved monomodal problems of the QMLTP library (**out of 580 problem**)
 (System versions: f2p-MSPASS 3.0, MleanSeP 1.2, LEO-II 1.3.2-M1.0, Satallax 2.2-M1.0, MleanTAP 1.3, MleanCoP 1.2.)

Logic/ Domain	ATP system					
	f2p-MSPASS	MleanSeP	LEO-II	Satallax	MleanTAP	MleanCoP
K/varying	-	-	72	104	-	-
K/cumul.	70	121	89	122	-	-
K/constant	67	124	120	146	-	-
D/varying	-	-	81	113	100	179
D/cumul.	79	130	100	133	120	200
D/constant	76	134	135	160	135	217
T/varying	-	-	120	170	138	224
T/cumul.	105	163	139	192	160	249
T/constant	95	166	173	213	175	269
S4/varying	-	-	140	207	169	274
S4/cumul.	121	197	166	238	205	338
S4/constant	111	197	200	261	220	352
S5/varying	-	-	169	248	219	359
S5/cumul.	140	-	215	297	272	438
S5/constant	131	-	237	305	272	438

Results for 20 multimodal logic problems: **LEO-II 15**, **Satallax 14**

- ▶ descriptions of 5 sound FML ATP systems in one talk (!)
- ▶ used QMLTP library for first evaluation
- ▶ one older system excluded because of soundness issues
- ▶ strongest ATPs: MleanCoP followed by Satallax
- ▶ best coverage: HOL approach

Future work includes:

- ▶ improvements of FML ATP systems
- ▶ extensions of QMLTP library and infrastructure

Experiments using the QMLTP Library

Logic/ Domain	ATP system					
	f2p-MSPASS	MleanSeP	LEO-II	Satallax	MleanTAP	MleanCoP
K/varying	-	-	0/529	165/356	-	-
K/cumul.	88/363	4/471	0/511	50/349	-	-
K/constant	42/405	2/471	12/481	45/328	-	-
D/varying	-	-	0/519	0/477	0/492	293/173
D/cumul.	33/407	0/461	0/500	0/464	0/472	194/171
D/constant	33/411	0/462	2/466	0/425	0/456	167/169
T/varying	-	-	0/478	30/320	0/453	121/223
T/cumul.	6/400	0/427	2/456	4/310	0/430	76/217
T/constant	6/410	0/428	2/427	1/295	0/415	66/213
S4/varying	-	-	0/458	30/289	1/421	109/199
S4/cumul.	0/433	0/397	0/430	6/270	1/384	115/163
S4/constant	0/448	0/401	2/397	4/255	1/368	100/162
S5/varying	-	-	0/427	27/265	1/369	132/148
S5/cumul.	0/418	-	0/379	0/244	1/315	126/118
S5/constant	0/436	-	2/359	0/231	1/315	116/118

The column entries x/y in this table show (i) the number x of problems that were *exclusively* solved (i.e. proved or refuted) by an ATP system in a particular logic&domain and (ii) the average CPU time y in seconds needed by an ATP system for solving all problems in a particular logic&domain (the full 600s timeout was counted for each failing attempt).